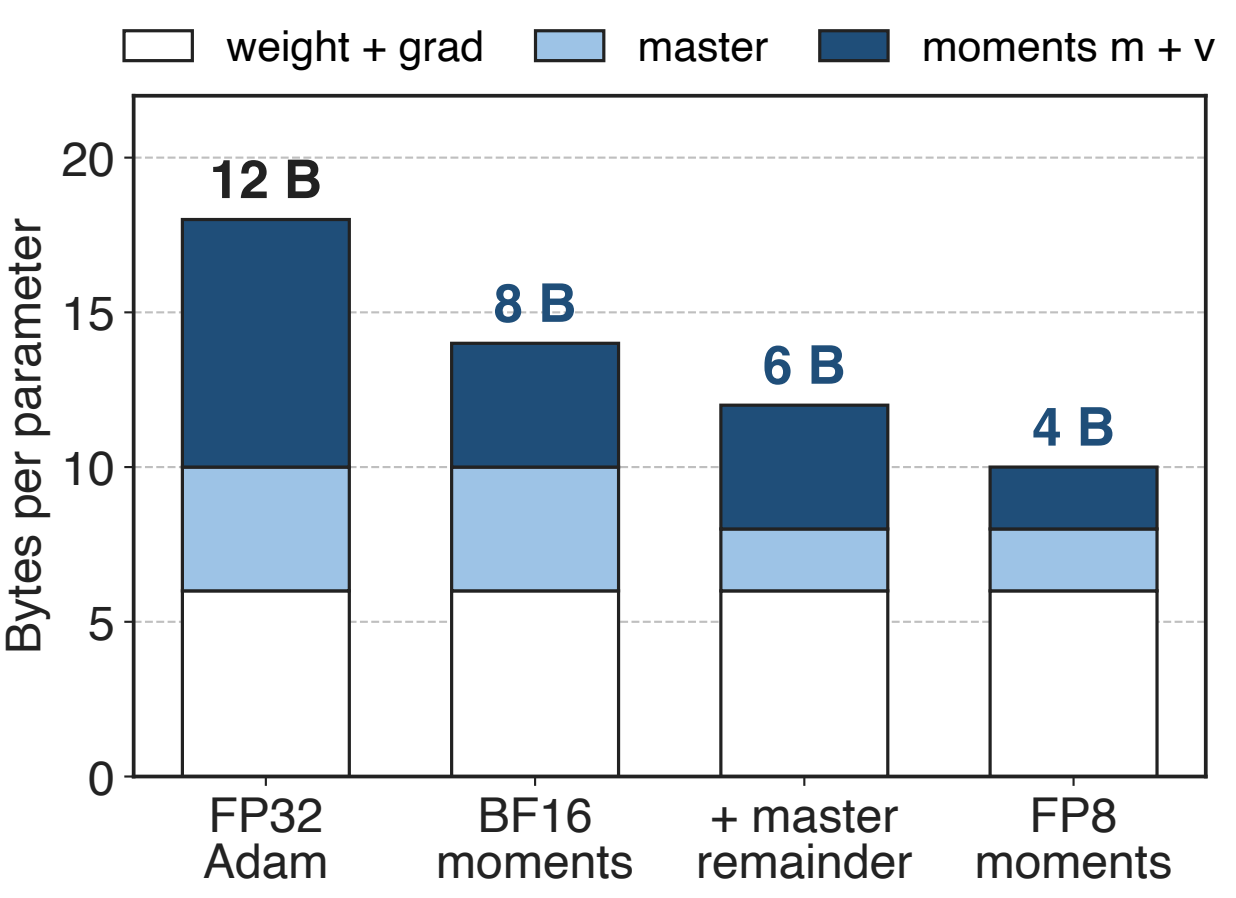
weight + grad
master
moments m + v
Bytes per parameter
0
5
10
15
20
12 B
8 B
6 B
4 B
FP32 Adam
BF16 moments
+ master remainder
FP8 moments